Progression of geographic atrophy in age-related macular degeneration imaged with spectral domain optical coherence tomography.

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To determine the area and enlargement rate (ER) of geographic atrophy (GA) in patients with age-related macular degeneration (AMD) using the spectral domain optical coherence tomography (SD-OCT) fundus image. Prospective, longitudinal, natural history study. Eighty-six eyes of 64 patients with ≥6 months of follow-up. Patients with GA secondary to AMD were enrolled in this study. Macular scans were performed using the Cirrus SD-OCT (Carl Zeiss Meditec, Dublin, CA). The areas of GA identified on the SD-OCT fundus images were quantified using a digitizing tablet. Reproducibility of these measurements was assessed and the ER of GA was calculated. The usefulness of performing square root transformations of the lesion area measurements was explored. Enlargement rate of GA. At baseline, 27% of eyes had a single area of GA. The mean total area at baseline was 4.59 mm(2) (1.8 disc areas [DA]). The mean follow-up time was 1.24 years. Reproducibility, as assessed with the intraclass correlation coefficient (ICC), was excellent on both the original area scale (ICC = 0.995) and the square root scale (ICC = 0.996). Intergrader differences were not an important source of variability in lesion size measurement (ICC = 0.999, 0.997). On average, the ER of GA per year was 1.2 mm(2) (0.47 DA; range, 0.01-3.62 mm(2)/year). The ER correlated with the initial area of GA (r = 0.45; P<0.001), but there were variable growth rates for any given baseline area. When the square root transformation of the lesion area measurements was used as a measure of lesion size, the ER (0.28 mm/yr) was not correlated with baseline size (r = -0.09; P = 0.40). In this cohort of lesions, no correlation was found between ER and length of follow-up. Square root transformation of the data helped to facilitate sample size estimates for controlled clinical trials involving GA. The SD-OCT fundus image can be used to visualize and quantify GA. Advantages of this approach include the convenience and assurance of using a single imaging technique that permits simultaneous visualization of GA along with the loss of photoreceptors and the retinal pigment epithelium that should correlate with the loss of visual function.

Progression of Geographic Atrophy with Subsequent Exudative Neovascular Disease in Age-Related Macular Degeneration: AREDS2 Report 24.

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To examine whether the rate of geographic atrophy (GA) enlargement is influenced by subsequent exudative neovascular age-related macular degeneration (nAMD) and hence, to explore indirectly whether nonexudative nAMD may slow GA enlargement. Post hoc analysis of a controlled clinical trial cohort. Age-Related Eye Disease Study 2 participants 50 to 85 years of age. Baseline and annual stereoscopic color fundus photographs were evaluated for (1) GA presence and area and (2) exudative nAMD presence. Two cohorts were constructed: eyes with GA at study baseline (prevalent cohort) and eyes in which GA developed during follow-up (incident cohort). Mixed-model regression of the square root of GA area was performed according to the presence or absence of subsequent exudative nAMD. Change over time in square root of GA area. Of the 757 eyes in the incident GA cohort, over a mean follow-up of 2.3 years (standard deviation [SD], 1.2 years), 73 eyes (9.6%) demonstrated subsequent exudative nAMD. Geographic atrophy enlargement in these eyes was significantly slower (0.20 mm/year; 95% confidence interval [CI], 0.12-0.28 mm/year) compared with the other 684 eyes in which subsequent exudative nAMD did not develop (0.29 mm/year; 95% CI, 0.27-0.30 mm/year; P = 0.037). Of the 456 eyes in the prevalent GA cohort, over a mean follow-up of 4.1 years (SD, 1.4 years), 63 eyes (13.8%)

demonstrated subsequent exudative nAMD. Geographic atrophy enlargement in these eyes was similar (0.31 mm/year; 95% CI, 0.24-0.37 mm/year) compared with the other 393 eyes in which subsequent exudative nAMD did not develop (0.28 mm/year; 95% CI, 0.26-0.29 mm/year; P = 0.37). In eyes with recent GA, GA enlargement before the development of exudative nAMD seems slowed. This association was not observed in eyes with more long-standing GA, which have larger lesion sizes. Hence, perilesional nonexudative choroidal neovascular tissue (presumably present before the development of clinically apparent exudation) may slow enlargement of smaller GA lesions through improved perfusion. This hypothesis warrants further evaluation in prospective studies.

Geographic atrophy progression secondary to age-related macular degeneration: Five years of follow-up.

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PurposeTo study the progression of geographic atrophy (GA) secondary to age-related macular degeneration over a five-year follow-up.MethodsEyes with GA included to assess demographic data, yearly optical coherence tomography (OCT) findings and the GA growth rate on infra-red (IR) images.ResultsA total of 41 eyes of 29 patients were included with a mean age of 81.76 ± 6.37 at baseline, and 65.51% were females. Over five years, there was a significant increase in the mean GA area from 8.44 ± 8.98 mm² to 13.32 ± 10.07 mm² ($P < 0.001$), with an annual growth rate of 1.14 ± 0.78 mm². The annual growth rates in females were slightly higher compared to males (1.29 ± 0.89 mm2 vs 0.96 ± 0.49 mm2, $p = 0.569$), and in smokers was slightly higher than non-smokers (1.35 ± 0.85 mm2 vs 0.94 ± 0.66 mm2, $p = 0.100$). Larger GA areas at the baseline showed higher GA progression in mm2 per year ($P = 0.04$). Smaller GA areas and fovea-spared GA at the baseline exhibited a larger percentage increase ($P < 0.001$ and $P = 0.015$, respectively). There was a lower GA progression rate in eyes with outer retinal tubulations (ORT) ($P = 0.027$), yet no significant correlation was found between GA progression and other OCT features.ConclusionsSmaller, fovea-sparing GA eyes experienced a more substantial proportional increase over five years. Also, The presence of ORT was associated with a slower rate of GA progression. Additionally, we observed a trend of faster GA growth in smokers and female genders.

Expert Consensus on Geographic Atrophy in the EU: A Call for Urgent Policy Action.

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Geographic atrophy is an eye disease that greatly interferes with the daily lives of patients and their families, posing a serious threat to the aging European demographic. Over the past 30 months, this initiative has assembled leading experts in the field of ophthalmology to share insights on the necessary policy steps that need to be taken to overcome this challenge on an EU-wide scale. Through analyzing best practices in Germany, Italy, France, and Spain, this consensus paper sets out a series of policy recommendations, which, if implemented, could greatly benefit all individuals affected by geographic atrophy. Amongst other features, these countries have provided valuable examples of awareness campaigns and an overall commitment to inclusive and comprehensive policies. The policy recommendations emerging from this paper include the adoption of comprehensive screening programs, retinal disease screening in the EU Driving License Directive, the development of a white paper at the

European Commission, and the creation of Council recommendations on eye health screening. Given the significant improvements made at the national level throughout the EU, countries will require unitary support at the European level to further develop their policies and successfully address the burden of geographic atrophy.

Diagnosis and Management of Patients With Geographic Atrophy Secondary to Age-Related Macular Degeneration: A Delphi Consensus Exercise.

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Geographic atrophy (GA) is a progressive and irreversible retinal disease with no comprehensive recommendations for diagnosis or monitoring. We used a Delphi approach to determine consensus in key areas around diagnosis and management of GA. A steering committee of eight retina specialists developed two sequential online surveys administered to eye care professionals (ECPs). Consensus was defined as agreement by ≥ 75% of respondents. Up to 177 ECPs from eight countries completed one or both surveys. Consensus was achieved in several topics related to diagnostic imaging, including the use of optical coherence tomography, and the urgent need for treatments and beneficial interventions to reduce the associated burden. Currently, low-vision aids and smoking cessation are considered the most beneficial interventions. We demonstrate consensus for diagnosis and management of patients with GA including best practices in patient identification and monitoring, and unmet needs. [Ophthalmic Surg Lasers Imaging Retina 2023;54:589-598.].

Patients with a fast progression profile in geographic atrophy have increased CD200 expression on circulating monocytes.

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Geographic atrophy (GA) is a progressing atrophy of the neuroretina with no treatment option. Age-related malfunction of retinal microglia amplifies response towards age-related tissue stress in age-related macular degeneration. Here, we investigated monocyte CD200 expression - the circulating middleman negotiating retinal microglial activity - in a poorly understood subtype of age-related macular degeneration. Prospective case-control study. Forty-six patients with GA and 26 healthy controls were included. All participants were subjected to a structured interview and detailed retinal examination. Controls were recruited from patient's spouses accompanying them in the clinic to match the groups best possibly. Participants had no history of immune disorders or cancer, and did not receive any immune-modulating medication. Patients did not have any history or sign of choroidal neovascularization in either eye. Fresh drawn blood was stained with monoclonal antibodies and prepared for flow cytometry to evaluate CD200 expression in monocytes and their functional subsets. The percentage of CD200+ monocytes in patients and controls. We found that monocytes were more CD200 positive in patients with GA compared to healthy age-matched controls. Then, we explored the potential relationship between CD200 expression and important fundus autofluorescence patterns that predict disease progression. Patients with a high risk of progression (patients with high degree of hyperautofluorescence) had distinctly increased CD200 expression compared to other patients with GA. Our data reveals that abnormal monocytic CD200 expression is present in GA, and in particular among those identified as fast progressors.

Change in area of geographic atrophy in the Age-Related Eye Disease Study: AREDS report number 26.

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To characterize progression of geographic atrophy (GA) associated with age-related macular degeneration in AREDS as measured by digitized fundus photographs. Fundus photographs from 181 of 4757 AREDS participants with a GA area of at least 0.5 disc areas at baseline or from participants who developed bilateral GA during follow-up were scanned, digitized, and evaluated longitudinally. Geographic atrophy area was determined using planimetry. Rates of progression from noncentral to central GA and of vision loss following development of central GA included the entire AREDS cohort. Median initial lesion size was 4.3 mm(2). Average change in digital area of GA from baseline was 2.03 mm(2) (standard error of the mean, 0.24 mm(2)) at 1 year, 3.78 mm(2) (0.24 mm(2)) at 2 years, 5.93 mm(2) (0.34 mm(2)) at 3 years, and 1.78 mm(2) (0.086 mm(2)) per year overall. Median time to developing central GA after any GA diagnosis was 2.5 years (95% confidence interval, 2.0-3.0). Average visual acuity decreased by 3.7 letters at first documentation of central GA, and by 22 letters at year 5. Growth of GA area can be reliably measured using standard fundus photographs that are digitized and subsequently graded at a reading center. Development of GA is associated with subsequent further growth of GA, development of central GA, and loss in central vision.

DrugnomeAI is an ensemble machine-learning framework for predicting druggability of candidate drug targets.

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The druggability of targets is a crucial consideration in drug target selection. Here, we adopt a stochastic semi-supervised ML framework to develop DrugnomeAI, which estimates the druggability likelihood for every protein-coding gene in the human exome. DrugnomeAI integrates gene-level properties from 15 sources resulting in 324 features. The tool generates exome-wide predictions based on labelled sets of known drug targets (median AUC: 0.97), highlighting features from protein-protein interaction networks as top predictors. DrugnomeAI provides generic as well as specialised models stratified by disease type or drug therapeutic modality. The top-ranking DrugnomeAI genes were significantly enriched for genes previously selected for clinical development programs (p value < 1 × 10-308) and for genes achieving genome-wide significance in phenome-wide association studies of 450 K UK Biobank exomes for binary (p value = 1.7 × 10-5) and quantitative traits (p value = 1.6 × 10-7). We accompany our method with a web application (http://drugnomeai.public.cgr.astrazeneca.com) to visualise the druggability predictions and the key features that define gene druggability, per disease type and modality.

Hijacked enhancer-promoter and silencer-promoter loops in cancer.

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Recent work has shown that besides inducing fusion genes, structural variations (SVs) can also contribute to oncogenesis by disrupting the three-dimensional genome organization and dysregulating gene expression. At the chromatin-loop level, SVs can relocate enhancers or silencers from their original genomic loci to activate oncogenes or repress tumor suppressor genes. On a larger scale, different types of alterations in topologically associating domains (TADs) have been reported in cancer, such as TAD expansion, shuffling, and SV-induced neo-TADs. Furthermore, the transformation from normal cells to cancerous cells is usually coupled with active or repressive compartmental switches, and cancer-specific compartments have been proposed. This review discusses the sites, and the other latest advances in studying how SVs disrupt higher-order genome structure in cancer, which in turn leads to oncogene dysregulation. We also highlight the clinical implications of these changes and the challenges ahead in this field.

Genome-wide detection of enhancer-hijacking events from chromatin interaction data in rearranged genomes.

PMID: 34092790 | Indexed in PubMed

Recent efforts have shown that structural variations (SVs) can disrupt three-dimensional genome organization and induce enhancer hijacking, yet no computational tools exist to identify such events from chromatin interaction data. Here, we develop NeoLoopFinder, a computational framework to identify the chromatin interactions induced by SVs, including interchromosomal translocations, large deletions and inversions. Our framework can automatically resolve complex SVs, reconstruct local Hi-C maps surrounding the breakpoints, normalize copy number variation and allele effects and predict chromatin loops induced by SVs. We applied NeoLoopFinder in Hi-C data from 50 cancer cell lines and primary tumors and identified tens of recurrent genes associated with enhancer hijacking. To experimentally validate NeoLoopFinder, we deleted the hijacked enhancers in prostate adenocarcinoma cells using CRISPR-Cas9, which significantly reduced expression of the target oncogene. In summary, NeoLoopFinder enables identification of critical oncogenic regulatory elements that can potentially reveal therapeutic targets.